

Key Terms in this Chapter

Term	Description
Absolute coordinates	Distance measured from a fixed reference point.
Aperture	Effective diameter of the cursor on the screen.
Cartesian coordinates	A rectangular system of measurement to locate points in the drawing area.
Construction line	A line used as a guide to create a framework for the drawing elements.
Isometric	A view of an object tilted at 30° angle on both sides.
Multi-line	A unified drawing element made up of a number of parallel lines that can be drawn just by indicating a starting point and end point.
Object snaps	A method for indicating point locations using existing drawing objects as a reference.
Origin point	The 0,0 location of the coordinate system.
Pattern	A graphical design that can be filled in a specified drawing area.
Polar coordinates	A system to locate the points using an angle and distance.
Polyline	A unified drawing element containing many drawing elements such as lines and arcs.
Prototype drawing	A template drawing that has a number of preset defaults.
Relative coordinates	Distances measured from the last point entered.
Spline	A continuous curved line.
User-defined coordinate system	A mode of measurement that allows the user to set up a customized coordinate system.

Components of a CADD Drawing

CADD drawings are created by drawing individual components (also called drawing objects or entities) of the drawing such as lines, arcs, dimensions and symbols. On a drawing board, you may not notice how many lines and arcs you need to complete a diagram, but with CADD you need to be specific. You need to specify exactly what you want to draw and how you want to draw it. Consider this scenario: drawing a square with rounded edges. Using a drawing board, you would quickly draw four lines and round off the edges with a template. Using CADD to draw the similar square, you will have to draw four lines and four arcs separately. You must specify where each line and arc starts, where it ends, the center point of the arc, the radius, etc.

Fig. 2.1 illustrates the basic drawing components of CADD. As shown, there are lines, arcs, circles, ellipses, symbols, text, dimensions pointers, polylines, borders and patterns. You can complete most of the drawings by using these components of CADD.

To draw any component of the drawing, you need to access a particular drawing command. There are separate commands to draw text, dimensions, patterns, symbols, and other elements of a drawing. The command can be entered by typing it in the command line, selecting it from the menu or by clicking on a specific tool button.

When you enter a command, the screen displays the prompts that guide you through the steps to take to complete a task. For example, when you enter the line command, the screen displays the prompt to enter the starting point and the end point of line. When you enter the arc command, the screen displays the prompt to enter the radius, center point, starting point and end point. Similarly, you need to follow the prompts to draw text, dimensions and symbols.

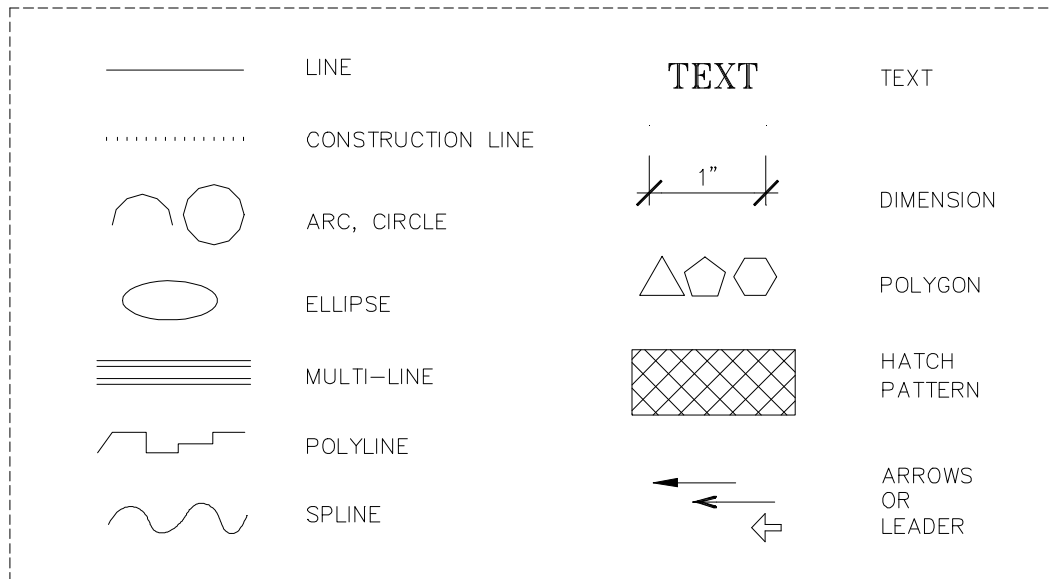


Fig. 2.1: *The components of a CADD drawing.*

Beginning to Draw

Here is a basic exercise using CADD. It is not taken from a specific CADD program. The objective of this exercise is to present an overall picture of how CADD works. It illustrates the concept of using different drawing functions. You don't need to sit in front of a computer to complete this exercise. Just read through the exercise to get a sense of procedures used in CAD.

In this exercise, a drawing tool bar is located on the left side of the screen. For illustration purposes, only the tool buttons used in this exercise are shown. A command window is located at the bottom of the screen that is used to enter commands as well as data. It also displays computer prompts when a specific function is entered using a tool button. The cursor is displayed like a pen that can be moved (with a pointing device) to make selections on the screen.

You will learn how to create a simple diagram using some lines, an arc, text and a dimension. A total of 20 screens (Fig.2.2 to Fig.2.6) illustrate the steps to complete the diagram.

To complete the exercise, the first task is to draw a grid (Fig. 2.2, Screen 1 & 2). The grid points will be used as a guide to draw the rest of the drawing elements. Let's begin...

Step	Action
1	Click on the grid tool button or enter the grid command in the command window.
2	Screen 2 shows the prompt to enter the grid spacing. Type 6" on the keyboard and then press the Return key. Grid is drawn to fill up the entire drawing window. <i>Note:</i> The number of grids displayed depends on the magnification level of the view, which can be adjusted to display appropriate number of grids.
3	Click on the line tool button and the line command is entered.
4	Screen 4 shows the prompt to enter the starting point of line. Place the cursor on grid referenced as "1" in the illustration and press the Enter button on the pointing device. <i>Note:</i> To enter the point exactly on the grid point, separate CADD functions called object snaps are used. These functions are discussed later in the chapter.

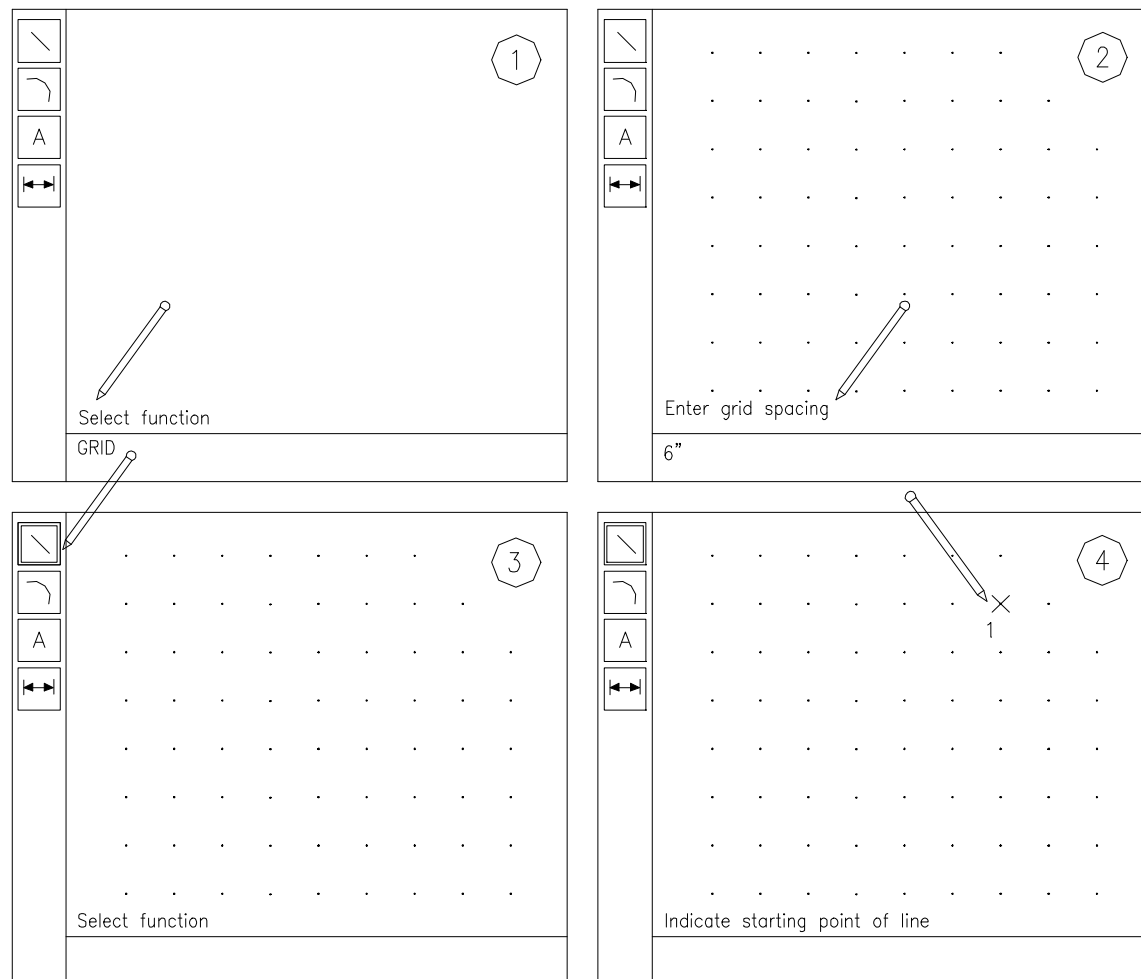


Fig. 2.2: Drawing a grid.

The following are the basic steps to draw lines and an arc (Fig. 2.3):

Step	Action
5	Screen 5 shows the prompt to enter the end point of line. Enter a point at the grid point 2 in the same manner as point 1 was entered. A line is drawn from 1 to 2.
6	Screen 6 shows the prompt to enter the end point of the next line. The previous point of line entered automatically becomes the starting point of next line. Continue to draw lines in the same manner from points 3 to 7 as shown. Note: The numbers shown in the illustration are for reference only.
7	Enter the arc command by clicking on the arc tool button.
8	Screen 8 shows the sub-menu with a number of methods for drawing arcs. Select the Center point & radius method. Note: Different methods of drawing arcs are discussed in Chapter 3 “The Drawing Tools.”

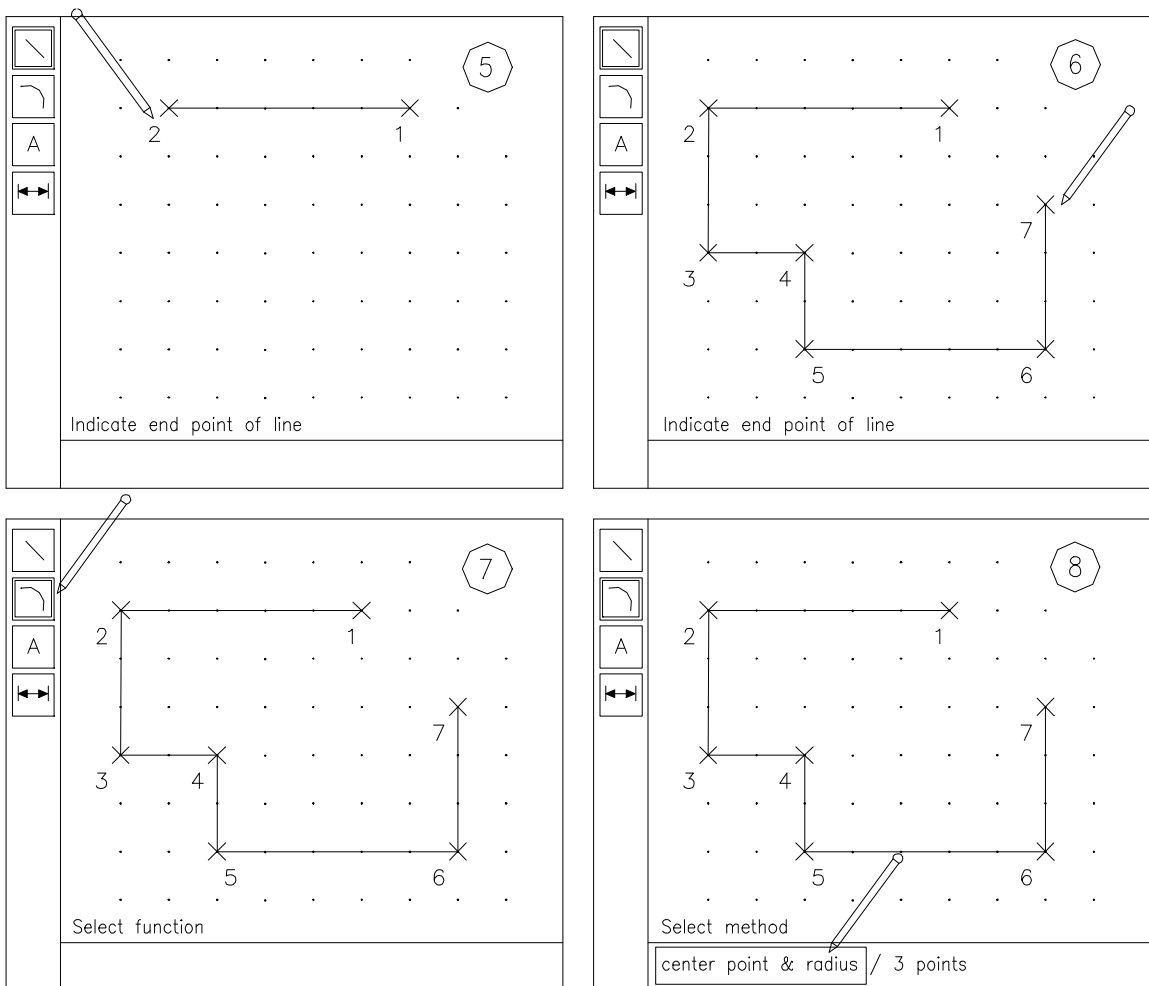


Fig. 2.3: Drawing lines.

... continuing to draw an arc (Fig. 2.4):

Step	Action
9	Screen 9 shows the prompt to enter the radius value of the arc. Enter 12" on the keyboard followed by the Return key.
10	Screen 10 shows the prompt to enter the center point of the arc. Enter the center point of the arc at grid point 8.
11	Screen 11 shows the prompt to enter the starting point of the arc. Enter the starting point of the arc at 7.
12	Screen 12 shows the prompt to enter the end point of the arc. Enter the end point of the arc at point 1. An arc is drawn as shown.

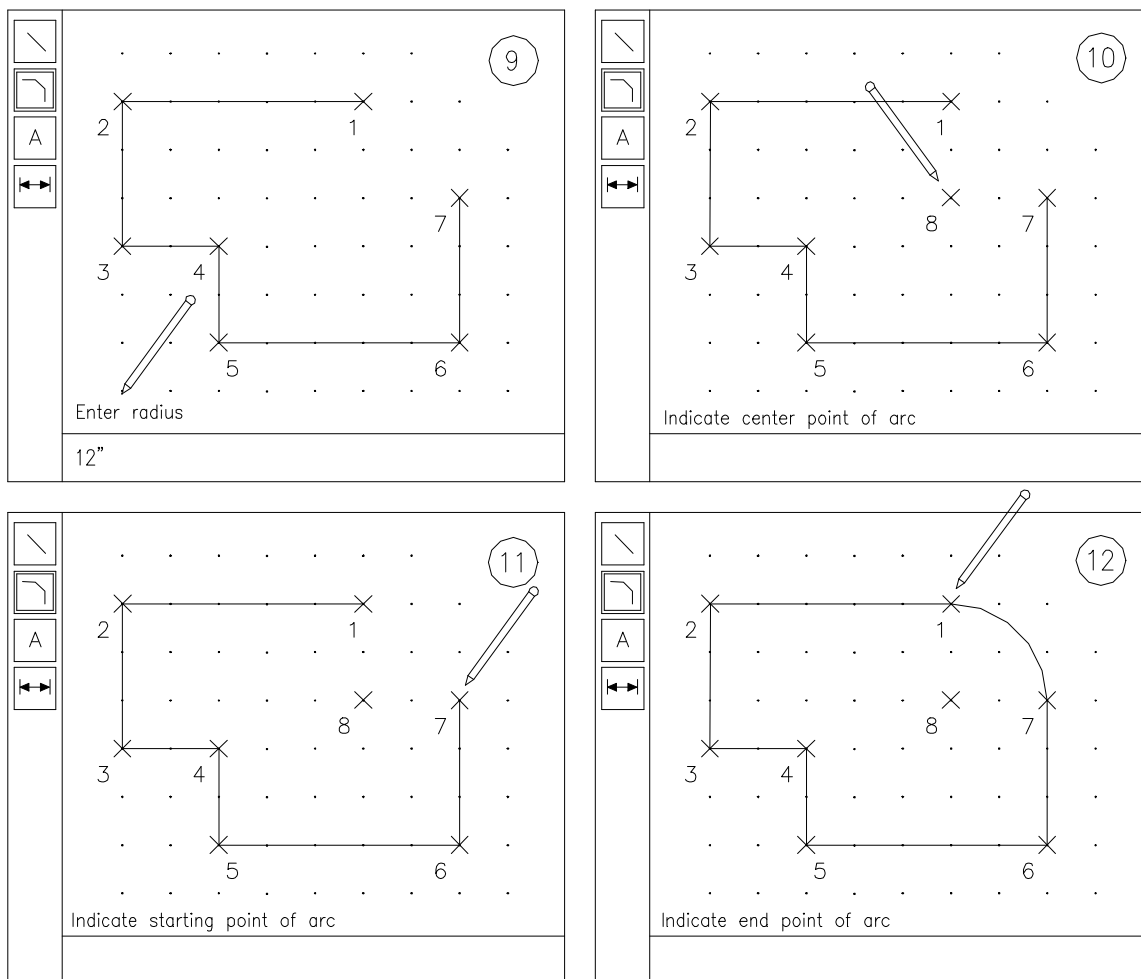


Fig. 2.4: Drawing an arc.

The following are the basic steps to write text (Fig. 2.5):

Step	Action
13	Click on the text tool button and the text drawing command is entered.
14	Screen 14 displays different methods of drawing text. Select the standard text option from the menu. Note: Different methods of drawing text are discussed in Chapter 3 "The drawing tools."
15	Screen 15 shows the prompt to enter text. Type the word PLAN and then press the Return key.
16	Screen 16 shows the prompt to enter the location of the text. Place the cursor approximately at the bottom of the diagram and press the Enter button on the mouse. The text is drawn as shown. Note: When writing text, it is generally not critical to place text at an exact location. You can locate it by approximation. However, if required, it can be placed at an exact position.

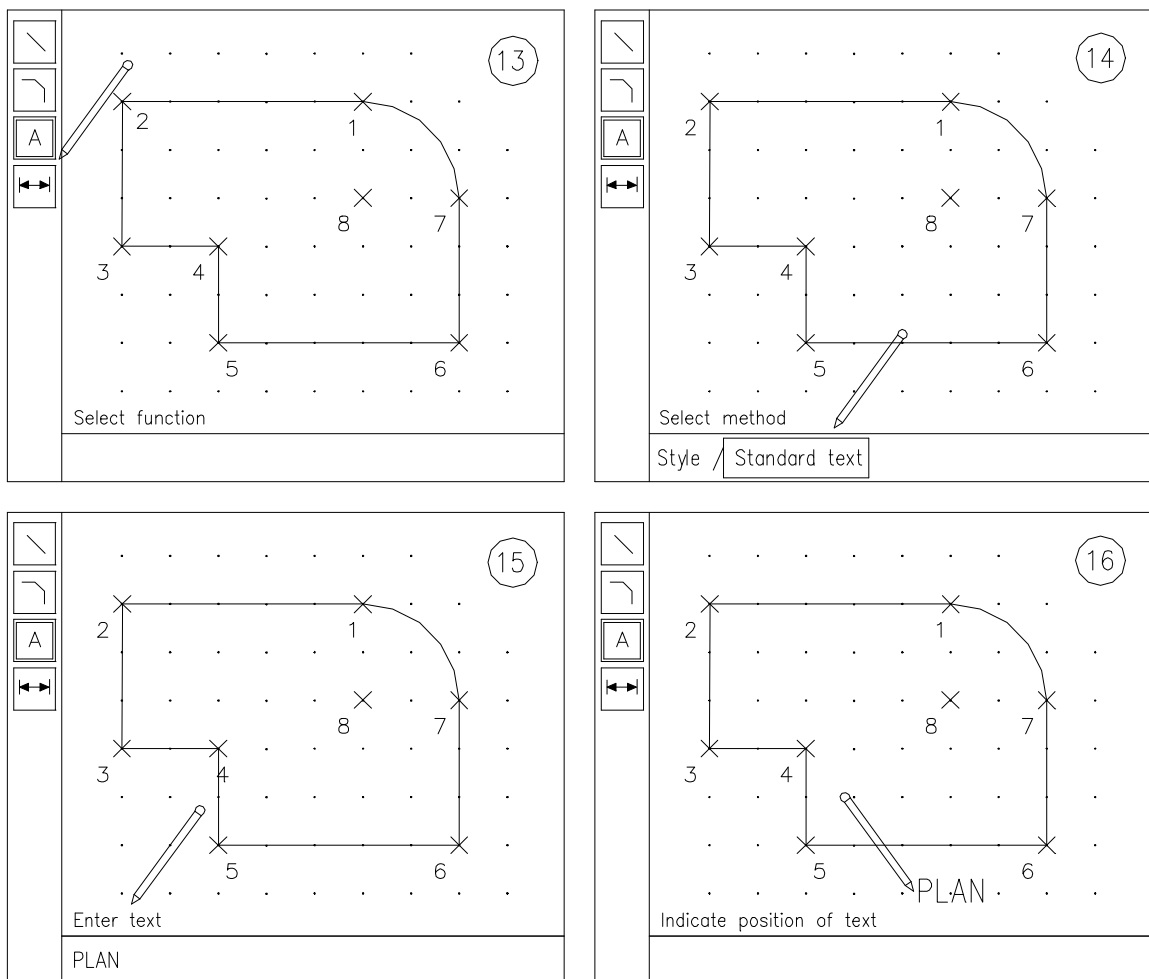


Fig. 2.5: Drawing text.

The following are the basic steps to draw dimension (Fig. 2.6):

Step	Action
17	Click on the dimension tool button and the dimension command is entered.
18	Screen 18 shows the sub-menu displaying methods for drawing dimensions. Select the Horizontal option from the sub-menu to draw a horizontal dimension.
19	Screen 19 shows the prompt to enter the first dimensioning point. Place the cursor on grid point 2 and press the Enter button on the cursor.
20	Screen 20 shows the prompt to enter second dimensioning point. Place the cursor on grid point 1 and press the Enter button on the cursor. A horizontal dimension is drawn as shown. The computer automatically calculates the dimension value and draws appropriate annotations required for the dimension.

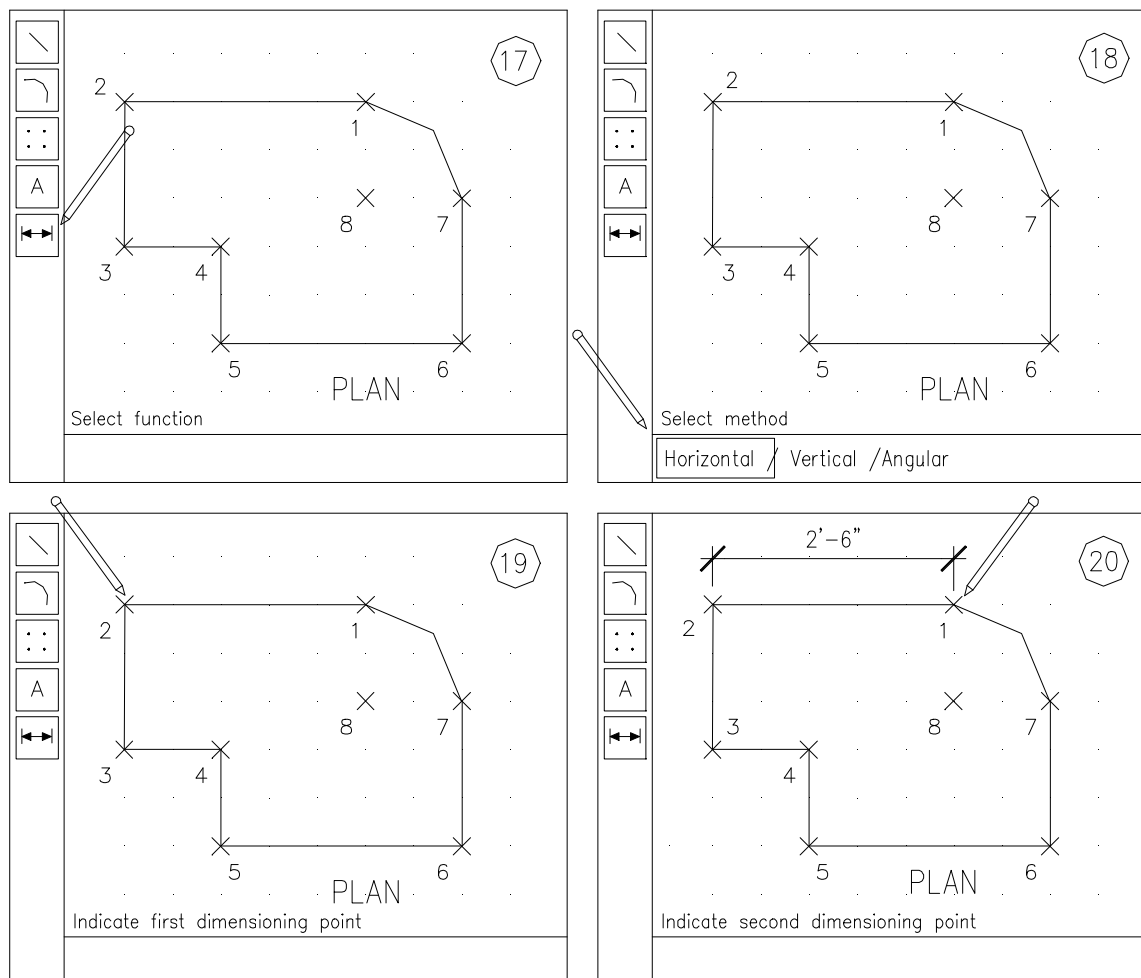


Fig. 2.6: Drawing dimension.

Notes:

- In the exercise above, we created nine components of a CADD drawing: six lines, one arc, one text and one dimension. All of these components maintain their own identity. For example, a dimension may be made up of lines and text, but the computer recognizes it as an independent entity. You can move it or edit it in a drawing only as a unified entity.
- CADD keeps track of all the components in a drawing and all the relative data entered to create them. Suppose you draw a 9-inch-long line and later decide to add a 6-inch line to it. Although it will look like one single line on-screen as well on paper, the computer will recognize them as two separate lines.

Locating Points Accurately

In the previous exercise you entered a number of points to draw different drawing elements. The points were entered using the grid points as a reference. However, you can not be certain that the points were entered exactly on the specified grid points. Even if the points were off by $1/100^{\text{th}}$ of an inch, this inaccuracy is not acceptable in a CADD drawing.

CADD includes separate functions that enable you to locate points with pinpoint accuracy. These are called object snaps that are included in a point-selection menu or toolbar. You can use a specific object snap tool button to locate a point at a specific point.

Fig. 2.7 shows an example of a point selection toolbar that shows a number of options to locate points. When you want to locate a point exactly at the end point of a line or arc, click on the end point tool button and then indicate a corresponding point in the drawing. The cursor just needs to be near the end point and it automatically snaps to it.

When you need to locate a point at the intersection of lines or arcs, you can use the intersection object snap. To locate a point exactly in the middle of a line, you can use the mid point object snap. Other options in the tool bar allow you to locate points perpendicular to a line, at the center point of a circle, snap to an existing point, etc.

Notes:

- Using object snaps is the key to creating professional CADD drawings. At times, you may feel that you quickly need to finish a drawing and you may save some time without using the object snaps. This will certainly result in an inaccurate drawing. When you dimension this drawing, it will create inaccurate dimensions.

- Most CADD programs allow you to set specific object snaps as a default. When a default is set, you don't need to click on the object snap tool button. The default is automatically used when you enter a point. For example, in the previous exercise, you can set existing point (or grid point) as a default and all the points will be entered exactly on the grid points.
- At times you will find that you are not able to locate a point when you have selected an object snap. It may be that you have selected an object snap, but did not point to the same condition in the drawing. If the computer cannot find the corresponding point, it displays an error message.

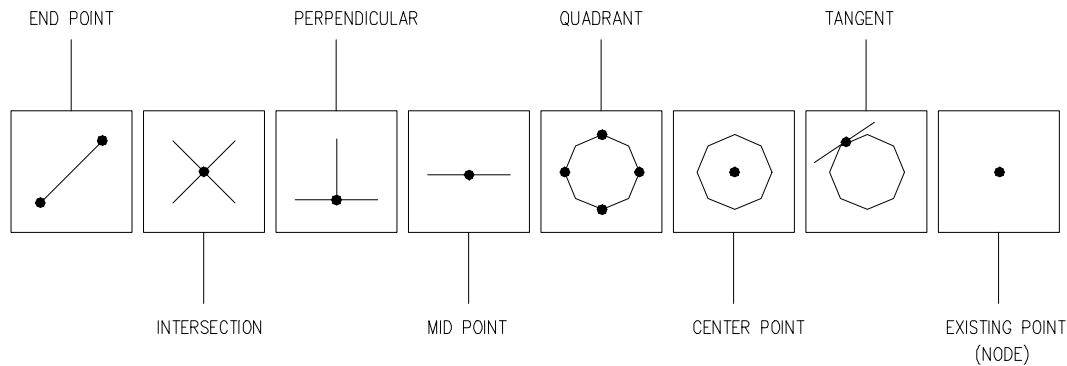


Fig. 2.7: Object snaps are used to precisely locate points in the drawing area.

The Coordinate System

The coordinate system is another method of locating points in the drawing area. It enables you to locate points by specifying distances from a fixed reference point. You can locate a point by giving its distance in the horizontal direction, vertical direction, measuring along an angle, etc.

The coordinate system is available when a function requires data input in the form of point locations. You may use it while drawing, editing or any time you need to locate a point. The most common coordinate systems are as follows:

- Cartesian coordinates
- Polar coordinates

Note:

The method of entering coordinates varies from one program to another. For example, in AutoCAD, the coordinates can be entered directly in the command window with the keyboard. In MicroStation and Cadkey, special dialog boxes let you enter the coordinates.

Cartesian Coordinates

Cartesian coordinates is a rectangular system of measurement that enables you to locate points with the help of horizontal and vertical coordinates. The horizontal values, called X-coordinates, are measured along the X-axis. The vertical values, called Y-coordinates, are measured along the Y-axis. The intersection of the X-AXIS and Y-axes is called the Origin Point, which represents the 0,0 location of the coordinate system (Fig. 2.8).

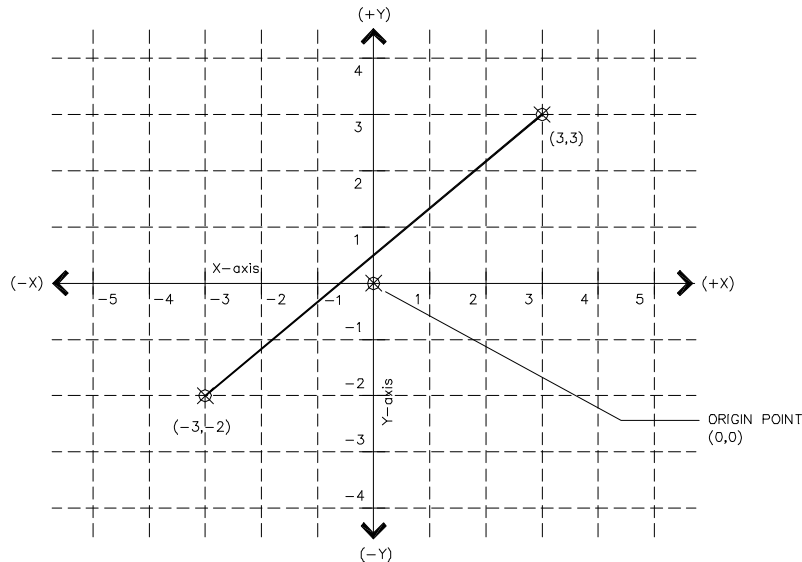


Fig. 2.8: *Measuring distances using Cartesian coordinates.*

The positive X-coordinates are measured to the right of origin point (or Y-axis) and the negative X-coordinates are measured to the left. Similarly, positive Y-coordinates are measured above the origin point (or X-axis) and the negative Y-coordinates are measured below. To enter a coordinate, you need to enter both the X and Y values separated by a comma (X, Y).

Example: To locate a point three units to the right and three units above the origin point, as shown in Fig. 2.8, enter X = 3, Y = 3 or (3,3). Similarly, to locate a point three units to the left and two units below, you need to enter X = -3, Y = -2 or (-3, -2). You can use fractions (if required) to enter the exact coordinate values.

Polar Coordinates

Polar coordinates allow you to define a point by specifying the distance and the direction from a given point. This mode of measurement is quite helpful when working with angles. To draw a line at an angle, you need to specify the length of the line (distance) and the angle.

Fig. 2.9 illustrates the concept of angle measurement in CADD. The radial lines show the angles measurements and the concentric circles show the distances. To draw a line as shown in the Fig. 2.9, you can enter two points using polar coordinates. To enter the point on the left, enter 120° as the angle and 5 units as the distance. To enter the point on the right, enter 30° as the angle and 4 units as the distance.

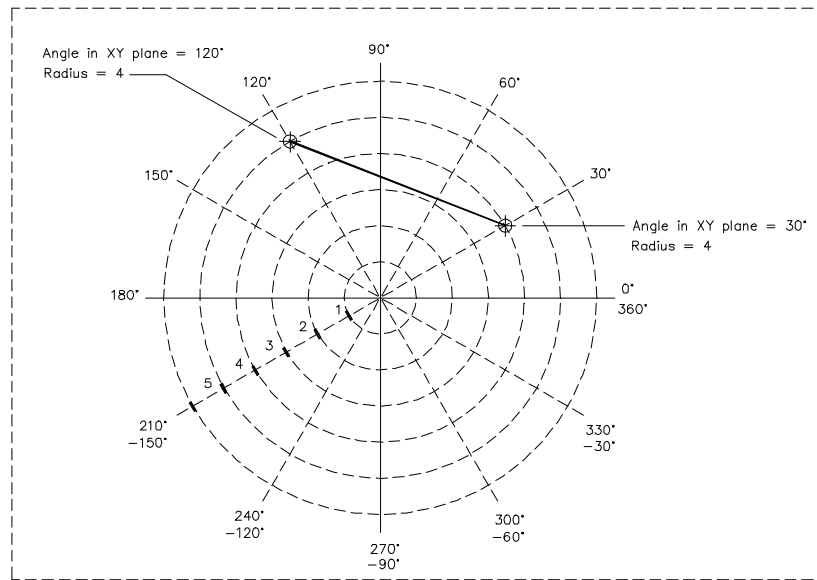


Fig. 2.9: *Measuring distances using polar coordinates.*

The Formats to Enter Coordinates

Cartesian or polar coordinate values can be entered in two formats:

- Absolute
- Relative

Absolute format is a way of measuring distances from a fixed reference location (origin point), which is the 0,0 location of the coordinate system. Consider this point to be stationary at all times. You can use this point as a reference to measure any distance in the drawing. Absolute coordinates are primarily used to adjust the alignment of diagrams in a drawing, to align one drawing with another or to make plotting adjustments.

Relative format is a way of measuring distances from the last point entered. All measurements are taken the same way as the absolute coordinates, with the exception that the relative coordinates are measured from the last point entered instead of the origin point. When a point is entered, it becomes the reference for entering the next point and so on. This mode of measurement is frequently used for drawing because it is always convenient to place the

drawing components relative to each other rather than a fixed reference point.

Note:

Each CADD program uses its own annotations to enter absolute or relative coordinates. The task table located at the end of this chapter shows some examples from leading CADD programs.

User-Defined Coordinate System

CADD allows you to create a user-defined coordinate system that can help simplify drawing. When you need to work with a complex drawing that has many odd angles you can create a user-defined coordinate system.

To define a user-defined coordinate system, you need to specify where you want the origin point and the direction of X and Y-axis. Thereafter, the computer works according to this customized coordinate system.

Let's say you need to draw or modify an odd-shaped diagram, such as the one shown in Fig. 2.10. It is very difficult to use cartesian or polar coordinates because they would involve extensive calculations. In this case, you can create a user-defined coordinate system that aligns with the odd angles of the diagram (28.5° and 118.5°). Now, if you draw a 10 units long line in the X direction by entering cartesian coordinates ($X=10, Y=0$), it will be automatically drawn at a 28.5° angle.

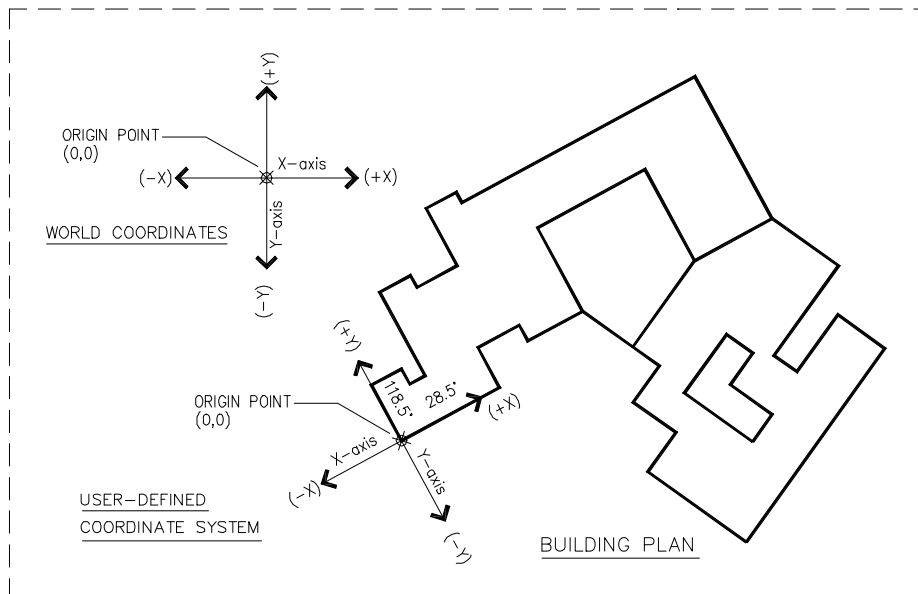


Fig. 2.10: A user-defined coordinate system can be used to work with odd-shaped diagrams.